

Elizabeth Janicek, EIT

ejanicek@wisc.edu | (815) 999-7923 | linkedin.com/in/eajanicek | 45 N Randall St, Madison WI, 53715

Portfolio Website: sites.google.com/view/ejanicek

EDUCATION

University of Wisconsin-Madison

Bachelor of Science: Mechanical Engineering expected May 2027; Certificate in Mathematics

GPA : 3.86 / 4.0

Engineer in Training Certification, California Board for Professional Engineers, License #188722 - August 2026

Member of Tau Beta Pi Engineering Honors Society; Dean's Honor List (6/6 Semesters)

ENGINEERING EXPERIENCE

Intro to Mechanical Engineering (ME 201), UW-Madison

Student Assistant

January 2024 - Present

- Executed iterative design methodology to develop 10 crane prototypes, 4 ramps, and documentation for 3 independent labs, enabling students to simulate core mechanical engineering principles on a weekly basis.
- Mentored 280 students each semester in advanced SolidWorks modeling, mechanical analysis, and material strengths, providing training in design principles, stress and force analysis, and mechatronic systems.

Busch Gardens Williamsburg, Williamsburg, VA

Leadership Intern Supply Chain

May 2026 - August 2026

- Managed warehouse operations, logistics, and inventory control for merchandise items as Assistant Supervisor.
- Led a capstone bakery redesign projecting a 244% ROI, covering a new kitchen floor layout, an NFPA 96 compliant fire suppression system, and 3D rendered front-of-house models, presented to senior leadership.

MasterMold LLC, Johnson Creek, WI

Engineering Intern

May 2025 - August 2025

- Designed and prototyped a robotic deburring mechanism with dual pneumatic air cylinders to replace an inconsistent manual process, eliminating faults and ensuring repeatability; produced SolidWorks drawings, part documentation, and a complete bill of materials for production.
- Developed a robot-to-press alignment tool using SolidWorks and FEA, creating a parametric base design deployed across 12 manufacturing units; prototyped via 3D printing, then machined in aluminum.

ENGINEERING LEADERSHIP

Badgers in Themed Entertainment (BiTE) - President and Founding Member

- Founded and scaled a registered engineering student organization in Spring 2024, recruiting 400+ members and building leadership infrastructure including SOPs, goal-tracking, budget, and a brand identity.
- Managed team projects such as a pneumatic launch coaster, model dark ride, and ride train, leading the team to the invitational round of TMU Thrill Design Competition in 2025 and 2026 for Mechanical Design.

ASTM F24 - NextGen Task Group Lead

- Organized NextGen Task Group programming for the February meeting with 91 student attendees, developing new member integration guides, coordinating networking activities, and leading task group meetings and panels.
- Initiated a NextGen Task Group project to document historical changes to all F24 standards, with the goal of conducting professional interviews and compiling the findings into an F24 standards guidebook.

Additional - Women in Mechanical Engineering (WME), Society of Women Engineers (SWE), ASME

ENGINEERING PROJECTS

Project Dark Ride - BiTE

- Designed and built 4 independent 1-DOF jellyfish animatronics for a model-scale interactive dark ride, incorporating laser-cut and 3D-printed components, and Arduino motor control programmed in Python.
- Engineered a 4-DOF octopus animatronic with 3D-printed living hinges for durable, life-like arm articulation surviving 20,000 cycles, researching print settings and material behavior to optimize rigid-plastic hinge performance.

Additional - See 6 other projects on portfolio website, including academic, professional, and personal work.

TECHNICAL SKILLS

Core Mechanical Engineering: SolidWorks, ANSYS, Blender, FEA analysis, MATLAB, EES (Engineering Equations Solver), Design for Manufacturability (DFM), ASTM Standards, Tolerance Analysis (GD&T), Six Sigma

Programming and Computer Science: Python, Java, JavaScript, HTML, SQL, C++, Simulink Scripting

Professional Communication and Design: Microsoft Suite, Google Suite, Adobe Illustrator, Presentational Speaking

Project Management: Scheduling, Design Review, Strategy, Risk Management, Gantt Chart